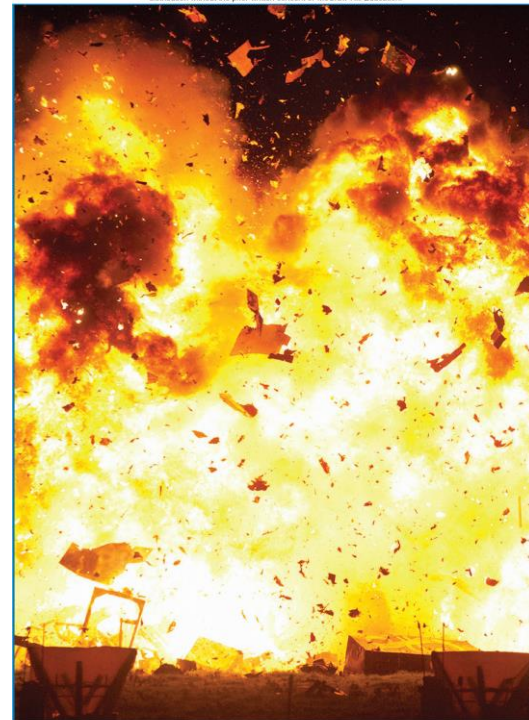


Thermochemistry

Chapter 6

- 6.1 The Nature of Energy and Types of Energy
- 6.2 Energy Changes in Chemical Reactions
- 6.3 Introduction to Thermodynamics
- 6.4 Enthalpy of Chemical Reactions
- 6.5 Calorimetry
- 6.6 Standard Enthalpy of Formation and Reaction
- 6.7 Heat of Solution and Dilution



Glan Cumming/Getty Images

Every **chemical reaction** obeys two fundamental laws:
the law of conservation of **mass** &
the law of conservation of **energy**

Energy

Energy is the capacity to do work.

- **Radiant energy** comes from the sun and is earth's primary energy source
- **Thermal energy** is the energy associated with the random motion of atoms and molecules
- **Chemical energy** is the energy stored within the bonds of chemical substances
- **Nuclear energy** is the energy stored within the collection of neutrons and protons in the atom
- **Potential energy** is the energy available by virtue of an object's position

All forms of energy can be converted (at least in principle) from one form to another.

Law of conservation of energy: the total quantity of energy in the universe is assumed constant.

Energy Changes in Chemical Reactions

Heat is the transfer of thermal energy between two bodies that are at different **temperatures**. 熱能

⇒ Heat ^{↓ 熱} flow ⇒ { heat absorbed 吸收
heat released 釋放

Temperature is a measure of the **thermal energy**.

Temperature ≠ Thermal Energy

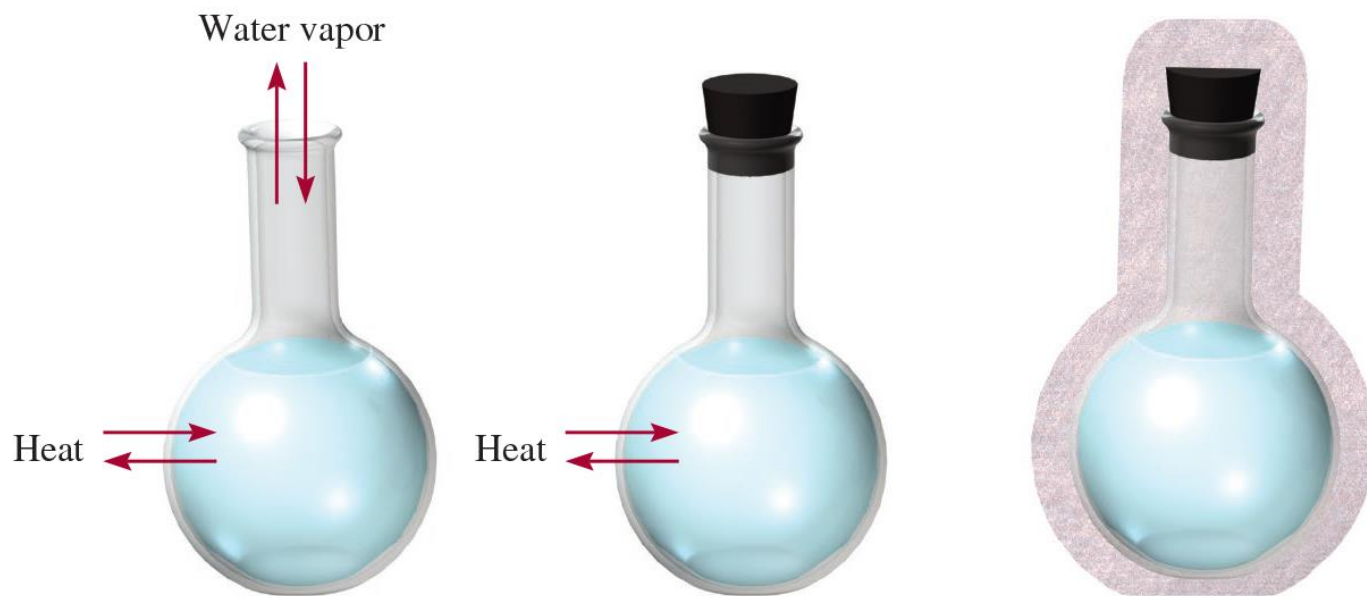


This infrared photo shows where energy (heat) leaks through the house. The more red the color, the more energy is lost to the outside.

Thermochemistry (1)

Thermochemistry is the study of heat change in chemical reactions. The **system** is the specific part of the universe that is of interest in the study.

Three systems represented by water in a flask: (a) an **open system**, which allows the exchange of both energy and mass with surroundings; (b) a **closed system**, which allows the exchange of energy but not mass; and (c) an **isolated system**, which allows neither energy nor mass to be exchanged (here the flask is enclosed by a vacuum jacket).



(a)

open

Exchange: mass & energy

(b)

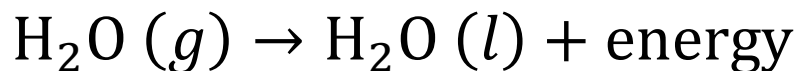
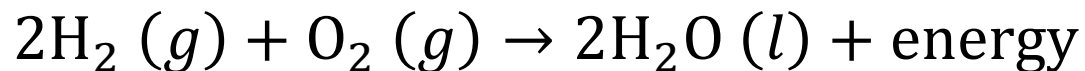
closed
energy

(c)

isolated
nothing

Exothermic and Endothermic Processes

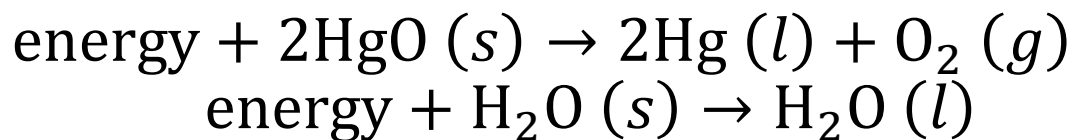
Exothermic process is any process that **gives off** heat – transfers thermal energy from the system to the surroundings.



We label the **reacting mixture (hydrogen, oxygen, and water molecules)** the **system** and the rest of the universe the **surroundings**.

The Hindenburg disaster. The Hindenburg, a German airship filled with **hydrogen gas**, was destroyed in a spectacular fire at Lakehurst, New Jersey, in 1937.

Endothermic process is any process in which heat has to be supplied to the system (HgO) from the surroundings.

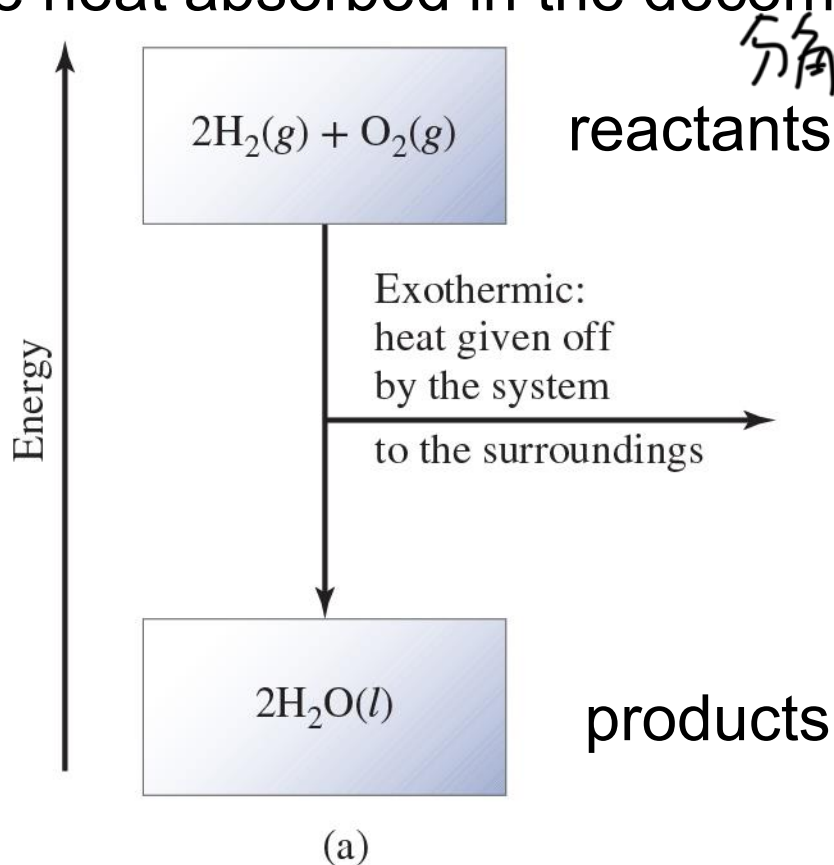


On heating, HgO decomposes to give Hg and O₂.

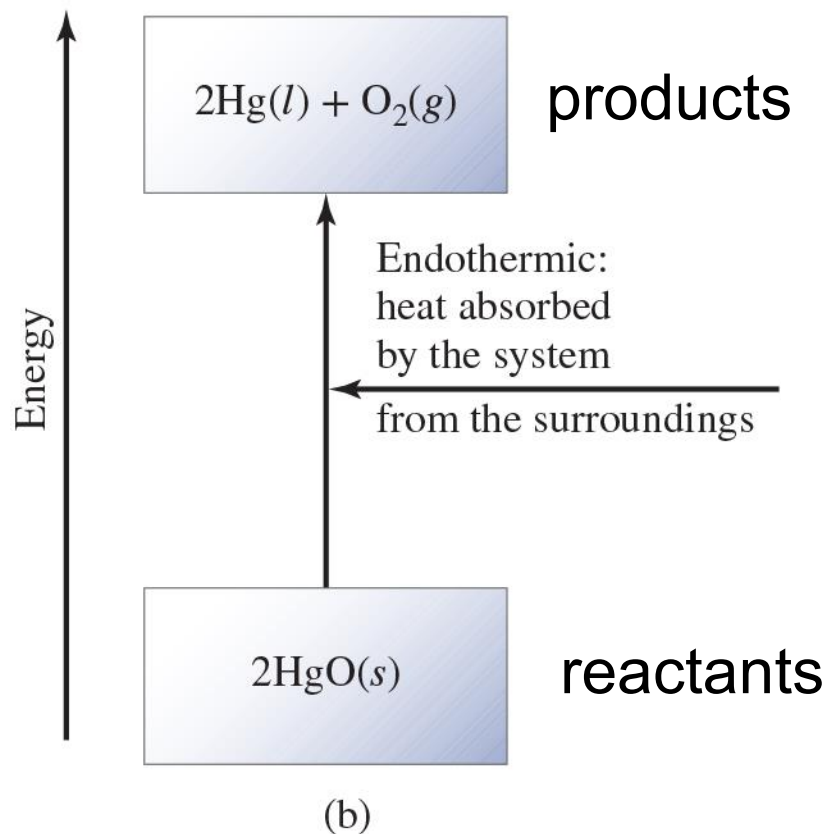
Schematic of Exothermic and Endothermic Processes

(a) An **exothermic** process. (b) An **endothermic** process.

Parts (a) and (b) are not drawn to the same scale; that is, the heat released in the formation of H_2O from H_2 and O_2 is not equal to the heat absorbed in the decomposition of HgO .



$$E_{\text{products}} < E_{\text{reactants}}$$



$$E_{\text{products}} > E_{\text{reactants}}$$

Thermodynamics

Thermodynamics is the scientific study of the interconversion of heat and other kinds of energy. **The first law of thermodynamics**
State functions are properties that are determined by the state of the system, regardless of how that condition was achieved.

⇒ **energy, pressure, volume, temperature**

When the state of a system changes, the magnitude of change in any **state function** depends only on the **initial and final states of the system** and not on how the change is accomplished.

$$\Delta U = U_{final} - U_{initial}$$

$$\Delta P = P_{final} - P_{initial}$$

$$\Delta V = V_{final} - V_{initial}$$

$$\Delta T = T_{final} - T_{initial}$$

Consider a gas at 2 atm, 300 K, and 1 L (the **initial state**).

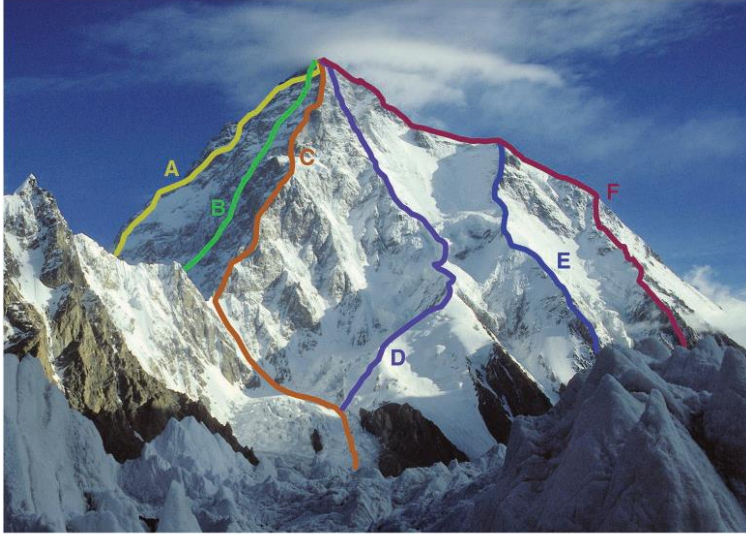
Suppose a process is carried out at **constant temperature** such that the **gas pressure decreases to 1 atm**. According to Boyle's law, its volume must increase to 2 L.

The **final state** then corresponds to 1 atm, 300 K, and 2 L.

The change in volume, $\Delta V = V_f - V_i = 2\text{L} - 1\text{L} = 1\text{L}$

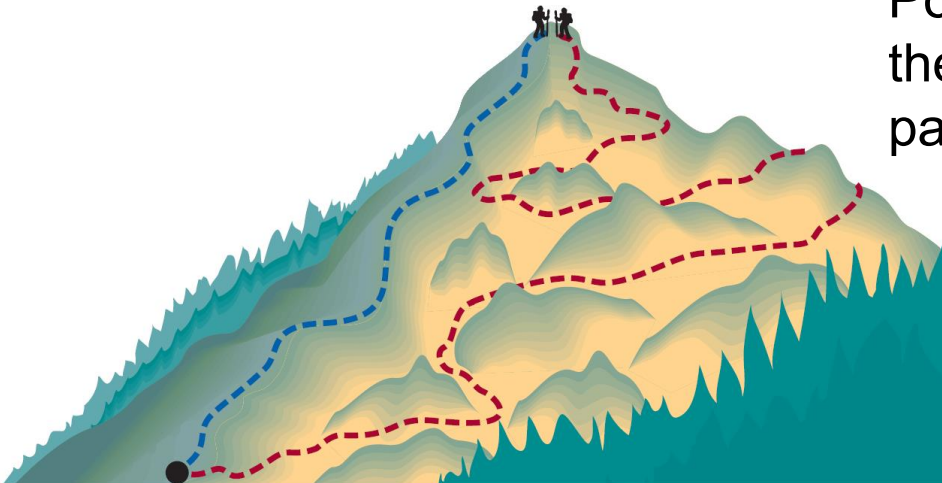
Examples

Copyright © McGraw-Hill Education. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill Education.



©Uwe Steffens/ullstein bild via Getty Images

The gain in gravitational potential energy that occurs when a person climbs from the base to the top of a mountain is independent of the path taken.



Potential energy of **hiker 1** and **hiker 2** is the same even though they took different paths.

Thermodynamics (1)

First law of thermodynamics – energy can be converted from one form to another, but cannot be created or destroyed.

$$U_{\text{system}} = U_{\text{kinetic energy}} + U_{\text{potential energy}}$$

$$U_{\text{kinetic energy}} = U_{\text{molecular motion}} + U_{\text{movement of electrons within molecules}}$$

$$U_{\text{potential energy in a molecule}} = U_{\text{electrons-nuclei}} + U_{\text{electrons-electrons}} + U_{\text{nuclei-nuclei}}$$



$$\Delta U = U_{\text{final}} - U_{\text{initial}} = U(\text{product}) - U(\text{reactants})$$

$$= \text{energy content of 1 mol SO}_2\text{(g)} - \text{energy content of [1 mol S(s) + 1 mol O}_2\text{(g)]}$$

this reaction gives off heat

$$\Rightarrow U(\text{product}) < U(\text{reactants})$$

$$\Rightarrow \Delta U < 0$$

Sulfur burning in air
to form SO₂

Interpreting the release of heat in this reaction to mean that some of the chemical energy contained in the molecules has been converted to thermal energy, we conclude that **the transfer of energy from the system to the surroundings does not change the total energy of the universe.**

Interpreting the release of heat in this reaction to mean that some of the chemical energy contained in the molecules has been converted to thermal energy, we conclude that **the transfer of energy from the system to the surroundings does not change the total energy of the universe.**

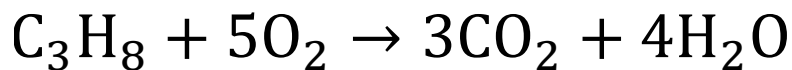
$$\Delta U_{sys} + \Delta U_{surr} = 0 \quad \text{or} \quad \Delta U_{sys} = -\Delta U_{surr}$$

If one system undergoes an energy change ΔU_{sys} , the rest of the universe, or the surroundings, must undergo a change in energy that is equal in magnitude but opposite in sign ($-\Delta U_{surr}$); energy gained in one place must have been lost somewhere else.

Copyright © McGraw-Hill Education. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill Education.



©Andrew Lambert Photography/SPL/Science Source



Exothermic chemical reaction!

Chemical energy **lost** by combustion = Energy **gained** by the surroundings
system surroundings

Another form of the **first law** for ΔU_{system}

$$\Delta U_{\text{sys}} = \Delta U = q + w$$

$\Delta U \equiv$ the change in internal energy of a system

$q \equiv$ the heat exchange between the system and the surroundings

$q \begin{cases} > 0: \text{endothermic process} \\ < 0: \text{exothermic process} \end{cases}$

$w \equiv$ the work done on (or by) the system

$w \begin{cases} > 0: \text{work done on the system by the surroundings} \\ < 0: \text{work done by the system on the surroundings} \end{cases}$

$w = -P\Delta V$ when a gas expands against a constant external pressure

Table 6.1 Sign Conventions for Work and Heat

Process	Sign
Work done by the system on the surroundings	-
Work done on the system by the surroundings	+
Heat absorbed by the system from the surroundings (endothermic process)	+
Heat absorbed by the surroundings from the system (exothermic process)	-

Another form of the **first law** for ΔU_{system}

$$\Delta U_{\text{sys}} = \Delta U = q + w$$

$\Delta U \equiv$ the change in internal energy of a system

$q \equiv$ the heat exchange between the system and the surroundings

$q \begin{cases} > 0: \text{endothermic process} \\ < 0: \text{exothermic process} \end{cases}$

$w \equiv$ the work done on (or by) the system

$w \begin{cases} > 0: \text{work done on the system by the surroundings} \\ < 0: \text{work done by the system on the surroundings} \end{cases}$

$w = -P\Delta V$ when a gas expands against a constant external pressure

If a system loses heat to the surroundings (exothermic) or does work on the surroundings, $\Delta U_{\text{sys}} < 0$. $\Rightarrow$ both $q < 0$ and $w < 0$.

If heat is added to the system (endothermic) or if work is done on the system, $\Delta U_{\text{sys}} > 0$. $\Rightarrow$ both $q > 0$ and $w > 0$.

Work Done on the System

Mechanical work—the **expansion or compression of a gas**.

Breathing and exhaling air involves the expansion and contraction of the tiny sacs called alveoli in the lungs.

The internal combustion engine of the automobile. The successive expansion and compression of the cylinders due to the combustion of the gasoline-air mixture provide power to the vehicle.

$$w = F \times d$$

$$P \times V = \frac{F}{d^2} \times d^3 = F \times d = w$$

The work done by the gas on the surroundings

$$w = -P \Delta V$$

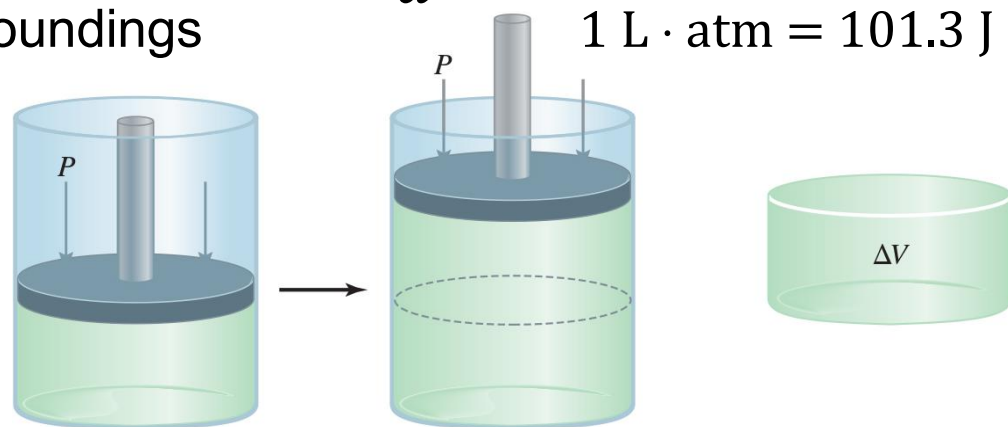
$$= -P (V_f - V_i)$$

If P is zero (the gas is expanding against a vacuum), $w=0$.

If P is some positive, nonzero value, then the work done is given by $-P\Delta V$.

(i) For gas expansion (work done by the system), $\Delta V > 0 \Rightarrow -P\Delta V < 0$.

(ii) For gas compression (work done on the system), $\Delta V < 0 \Rightarrow -P\Delta V > 0$



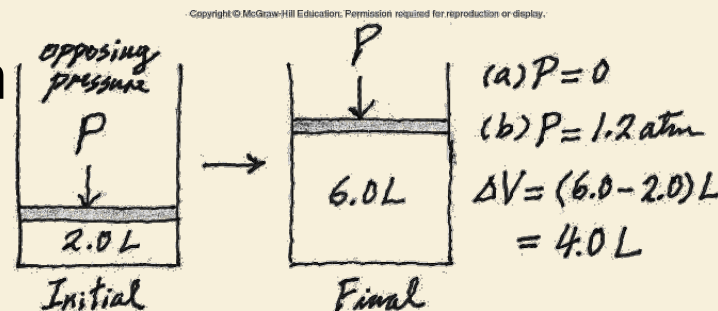
The expansion of a gas against a constant external pressure (such as atmospheric pressure). The gas is in a cylinder fitted with a weightless movable piston. The work done is given by $-P\Delta V$. Because $\Delta V > 0$, the work done is a negative quantity.

Example 6.1

A certain gas expands in volume from 2.0 L to 6.0 L at constant temperature. Calculate the work done by the gas if it expands

(a) against a vacuum

(b) against a constant pressure of 1.2 atm



Solution

(a) Because the external pressure is zero, no work is done in the expansion. $w = -P\Delta V = -(0)(6.0 - 2.0)\text{L} = 0$

(b) The external, opposing pressure is 1.2 atm, so

$$w = -P\Delta V = -(1.2\text{atm})(6.0 - 2.0)\text{L} = -4.8\text{L} \cdot \text{atm}$$

To convert the answer to joules, we write

$$w = -4.8\text{L} \cdot \text{atm} \times \frac{101.3\text{J}}{1\text{L} \cdot \text{atm}} = -4.9 \times 10^2 \text{J}$$

Example 6.1 shows that **work is not a state function**.

$$\Delta w \neq w_{\text{final}} - w_{\text{initial}}$$

Work done depends not only on the initial state and final state, but also on how the process is carried out, that is, on the path.

Heat

Heat (q) is not a state function—it takes **4184 J** of energy to raise the temperature of 100 g of water from **20 °C to 30 °C**.

This energy can be gained

(a) directly as heat energy from a Bunsen burner, without doing any work on the water;

(b) by doing work on the water without adding heat energy (for example, by stirring the water with a magnetic stir bar);

(c) by some combination of the procedures described in (a) & (b).

⇒ heat associated with a given process, like work, depends on how the process is carried out.

Regardless of which procedure is taken, the change in internal energy of the system, ΔU , depends on the sum of ($q + w$).

Example 6.2

The work done when a gas is compressed in a cylinder like that shown in Figure 6.5 is 462 J. During this process, there is a heat transfer of 128 J from the gas to the surroundings.

Calculate the energy change for this process.

Solution To calculate the energy change of the gas, we need Equation (6.1). Work of compression is positive and because heat is released by the gas, q is negative. Therefore, we have

$$\Delta U = q + w = -128 \text{ J} + 462 \text{ J} = 334 \text{ J}$$

As a result, the energy of the gas increases by 334 J.

Chemistry in Action: Making Snow and Inflating a Bicycle Tire (1st law of thermodynamics)

Making Snow

The secret of snowmaking is in the equation $\Delta U = q + w$.

A snowmaking machine contains a mixture of **compressed air** and **water vapor** at about **20 atm**.



Because of the large difference in **pressure between the tank and the outside atmosphere**, when the mixture is sprayed into the atmosphere it expands so rapidly that, as a good approximation, **no heat exchange occurs between the system (air and water) and its surroundings; $q \approx 0$. (adiabatic process)**

$$\Delta U = q + w = w$$

$$\Delta T < 0, \text{ SNOW!}$$

The system does work on the surroundings $\Rightarrow w < 0$
 $\Rightarrow$ a decrease in the system's energy.

$$\Delta U = C \Delta T$$

$$w < 0, \Delta U < 0$$

$$\overline{\text{KE}} = \frac{1}{2} m \overline{u^2} = CT$$

Chemistry in Action: Making Snow and Inflating a Bicycle Tire (1st law of thermodynamics)

$$U_{sys} = U_{KE} + U_{PE}$$

Average kinetic energy of a gas

$$U_{KE} = \overline{KE} = \frac{1}{2} m \overline{u^2} = CT \Rightarrow \Delta U_{KE} = C \Delta T$$

$$\Delta U_{PE} = \Delta U = C \Delta T$$

$$\therefore \Delta U < 0 \Rightarrow \Delta T < 0 \Rightarrow \text{SNOW!}$$

This cooling effect (or the decrease in the kinetic energy of the water molecules) is responsible for the formation of snow.

Inflating a Bicycle Tire

If you have ever pumped air into a bicycle tire, you probably noticed a warming effect at the valve stem.

The action of the pump compresses the air inside the pump and the tire. (The process is rapid enough to be treated as approximately adiabatic, so that $q = 0$ & $\Delta U = w$.)

$\therefore$ work is done on the gas (it is being compressed)

$$\Rightarrow w > 0, \text{ and } \Delta U > 0.$$

$\Rightarrow$ The temperature of the system increases

$$\Delta U = C \Delta T$$

Enthalpy of Chemical Reactions & 1st Law of Thermodynamics

Chemical reaction is run at constant volume, then $\Delta V = 0$

$$\Delta U = q + w = q - P\Delta V = q_v$$

Most chemical reactions occur under conditions of constant pressure (usually atmospheric pressure).

$$\Delta U = q + w = q_p - P\Delta V \Rightarrow q_p = \Delta U + P\Delta V$$

Enthalpy (H): $H = U + PV$

U , P , and V are all state functions, that is, the changes in $(U + PV)$ depend only on the initial and final states. $\Rightarrow H$ is a state function.

$$H = U + PV \Rightarrow \Delta H = \Delta U + \Delta(PV) = \Delta U + P\Delta(V) + V\Delta(P)$$

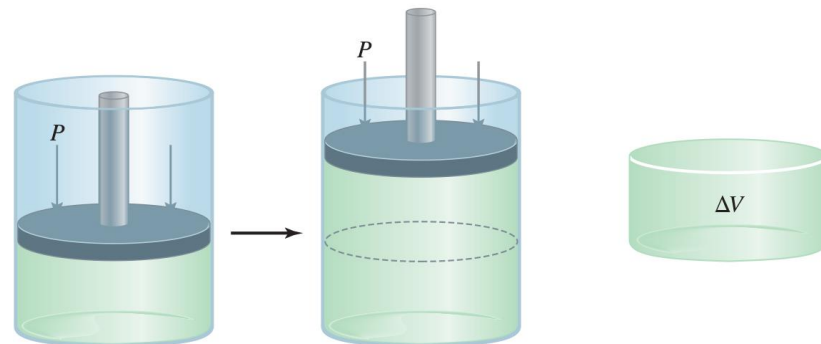
$$\text{i.e. } \Delta H = \Delta U + P\Delta V \quad (\Delta U = \Delta H - P\Delta V)$$

$$q_p = \Delta H$$

heat change

$$q_v = \Delta U; q_p = \Delta H$$

Copyright © McGraw-Hill Education. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill Education.



Enthalpy

Enthalpy (H) is used to quantify the **heat flow** into or out of a system in a process that occurs at **constant pressure**.

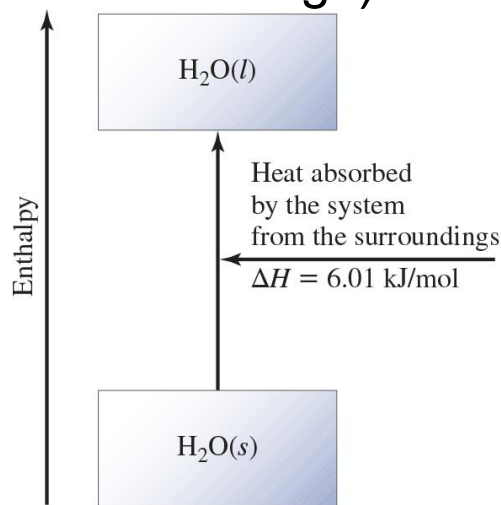
reactants $\rightarrow$ products

enthalpy of reaction, $\Delta H_{rxn} = \Delta H = H(\text{products}) - H(\text{reactants})$

ΔH = heat given off or absorbed during a reaction **at constant pressure**

endothermic process

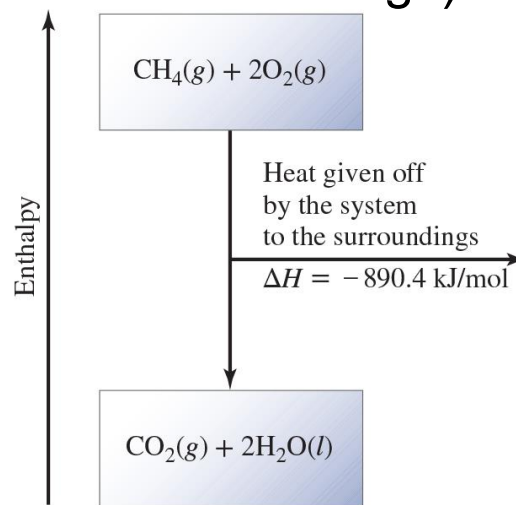
(heat **absorbed** by the system from the surroundings)



(a)

exothermic process

(heat **released** by the system to the surroundings)



(b)

$$H_{\text{products}} > H_{\text{reactants}} \Rightarrow \Delta H > 0 \quad H_{\text{products}} < H_{\text{reactants}} \Rightarrow \Delta H < 0$$

There is no way to know the enthalpies of individual products and reactants. In practice, we can only measure the difference in their values.

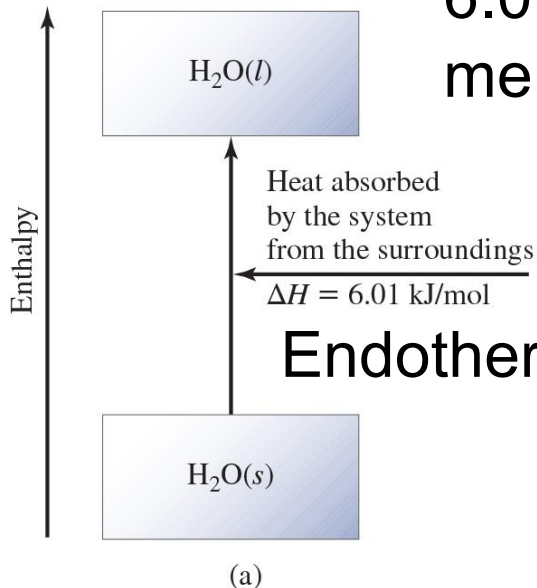
Thermochemical Equations (1)

6.01 kJ are absorbed for every **1 mole** of ice that melts at **0°C** and 1 atm.

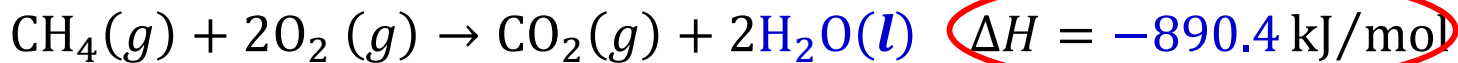


$\therefore$ pressure is constant, $q_p = \Delta H > 0$

Endothermic System absorbs heat

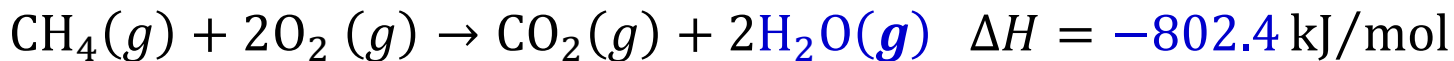
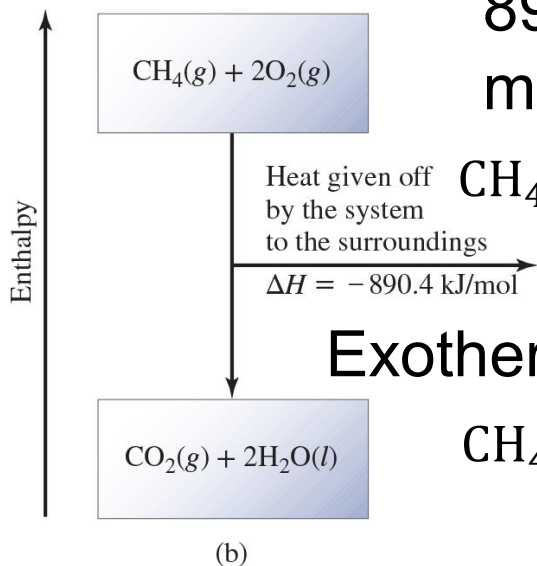


890.4 kJ are released for every **1 mole** of methane that is combusted at **25°C** and 1 atm.

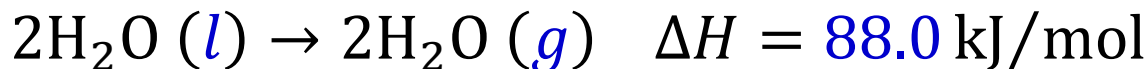
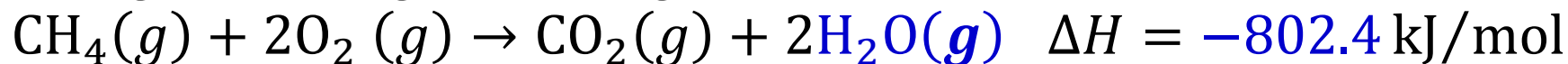
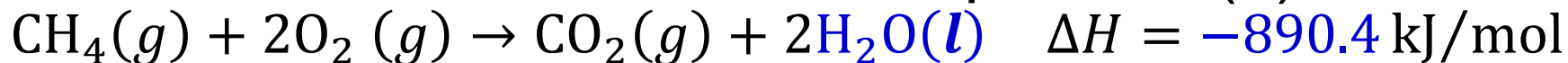


$\therefore$ pressure is constant, $q_p = \Delta H < 0$

Exothermic System gives off heat



Thermochemical Equations (2)



The stoichiometric coefficients always refer to the number of moles of a substance



If you reverse a reaction, the sign of ΔH changes



If you multiply both sides of the equation by a factor n , then ΔH must change by the same factor n .

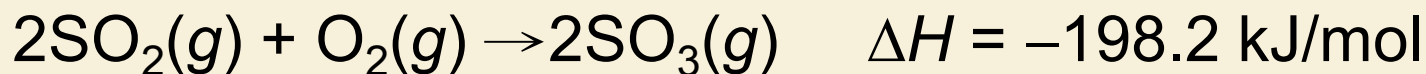


The physical states of all reactants and products must be specified in thermochemical equations.



Example 6.3

Given the thermochemical equation



calculate the heat evolved when 87.9 g of SO_2 (molar mass = 64.07 g/mol) is converted to SO_3 .

Solution We need to first calculate the number of moles of SO_2 in 87.9 g of the compound and then find the number of kilojoules produced from the exothermic reaction. The sequence of conversions is as follows:

gram of $\text{SO}_2 \rightarrow$ moles of $\text{SO}_2 \rightarrow$ kilojoules of heat generated

Therefore, the enthalpy change for this reaction is given by

$$\begin{aligned} \Delta H &= 87.9 \text{ g SO}_2 \times \frac{1 \text{ mol SO}_2}{64.07 \text{ g SO}_2} \times \frac{-198.2 \text{ kJ}}{2 \text{ mol SO}_2} \\ &= -136 \text{ kJ} \end{aligned}$$

and the heat released to the surroundings is 136 kJ.

A Comparison of ΔH and ΔU (1)



Constant pressure: $\Delta H = \Delta U + P\Delta V \Rightarrow \Delta U = \Delta H - P\Delta V$

Ignore the small change in the volume of the solution

At 25 °C, 1 mole $\text{H}_2 = 24.5 \text{ L}$ at 1 atm

$$P\Delta V = 1 \text{ atm} \times 24.5 \text{ L} = 2.5 \text{ kJ}$$

$$\Delta U = -367.5 \text{ kJ/mol} - 2.5 \text{ kJ/mol} = -370.0 \text{ kJ/mol}$$

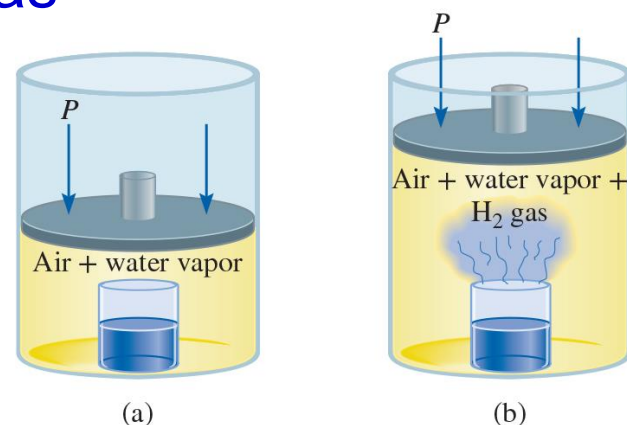


ΔU and ΔH are approximately the same. The reason ΔH is smaller than ΔU in magnitude is that some of the internal energy released is used to do gas expansion work, so less heat is evolved.

For reactions that do not involve gases,

ΔV is usually very small $\Rightarrow \Delta U \approx \Delta H$.

a) A beaker of water inside a cylinder fitted with a movable piston. The pressure inside is equal to the atmospheric pressure. (b) As the sodium metal reacts with water, the hydrogen gas generated pushes the piston upward (doing work on the surroundings) until the pressure inside is again equal to that of outside.

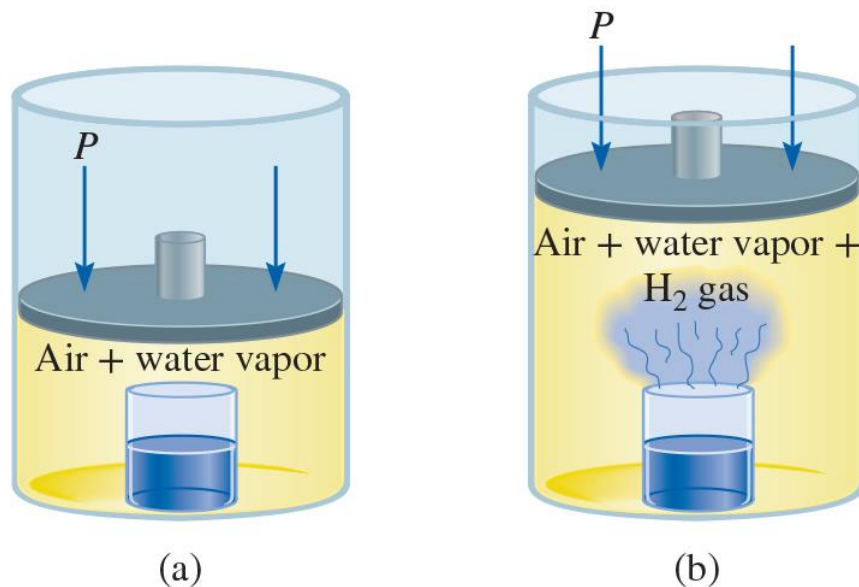


A Comparison of ΔH and ΔU

Constant pressure & ideal gas behavior:

$$H = U + PV \Rightarrow \Delta U = \Delta H - \Delta(PV) = \Delta H - \Delta(nRT) = \Delta H - RT\Delta n$$

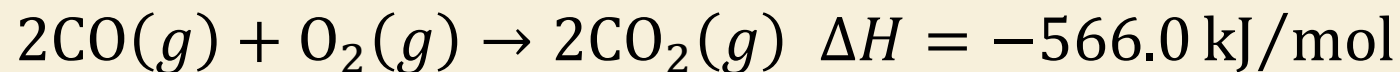
Δn = number of moles of **product** gases – number of moles of **reactant** gases



(a) A beaker of water inside a cylinder fitted with a movable piston. The pressure inside is equal to the atmospheric pressure. (b) As the sodium metal reacts with water, the hydrogen gas generated pushes the piston upward (doing work on the surroundings) until the pressure inside is again equal to that of outside.

Example 6.4

Calculate the change in internal energy when 2 moles of CO are converted to 2 moles of CO₂ at 1 atm and 25°C:



Solution

3 moles of gases are converted to 2 moles of gases so that

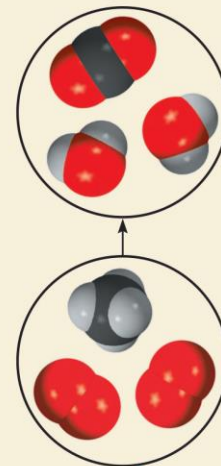
$$\Delta n = n_{\text{product gas}} - n_{\text{reactant gases}} = 2 - 3 = -1$$

Using $R = 8.314 \text{ J/K} \cdot \text{mol}$ and $T = 298 \text{ K}$

$$\Delta U = \Delta H - RT\Delta n$$

$$= -566.0 \text{ kJ/mol} - (8.314 \text{ J/K} \cdot \text{mol}) \left(\frac{1 \text{ kJ}}{1000 \text{ J}} \right) (298 \text{ K})(-1)$$

$$= -563.5 \text{ kJ/mol}$$



©McGraw-Hill Education/Stephen Frisch

**Carbon monoxide
burns in air to form
carbon dioxide**

Specific Heat and Heat Capacity

The **specific heat (s)** of a substance is the amount of heat (q) required to raise the temperature of **one gram** of the substance by **one degree Celsius**. Units: $\text{J/g}\cdot^{\circ}\text{C}$

The **heat capacity (C)** of a substance is the amount of heat (q) required to raise the temperature of **a given quantity (m)** of the substance by **one degree Celsius**. Units: $\text{J}/^{\circ}\text{C}$

$$C = ms$$

Specific heat is an **intensive** property, whereas **heat capacity** is an **extensive** property.

Table 6.2

The Specific Heats of
Some Common
Substances

Substances	Specific Heat ($\text{J/g}\cdot^{\circ}\text{C}$)
Al	0.900
Au	0.129
C (graphite)	0.720
C (diamond)	0.502
Cu	0.385
Fe	0.444
Hg	0.139
Pb	0.158
H_2O	0.184
$\text{C}_2\text{H}_5\text{OH}$ (ethanol)	2.46

<Example> The specific heat of water is $4.184 \text{ J/g}\cdot^{\circ}\text{C}$, and the heat capacity of 60.0 g of water is $(60.0 \text{ g})(4.184 \text{ J/g}\cdot^{\circ}\text{C}) = 251 \text{ J}/^{\circ}\text{C}$

Heat (q) absorbed or released:

$$q = ms\Delta t = C\Delta t \quad \Delta t = t_{\text{final}} - t_{\text{initial}}$$

$q > 0$: endothermic processes

$q < 0$: exothermic processes

Example 6.5

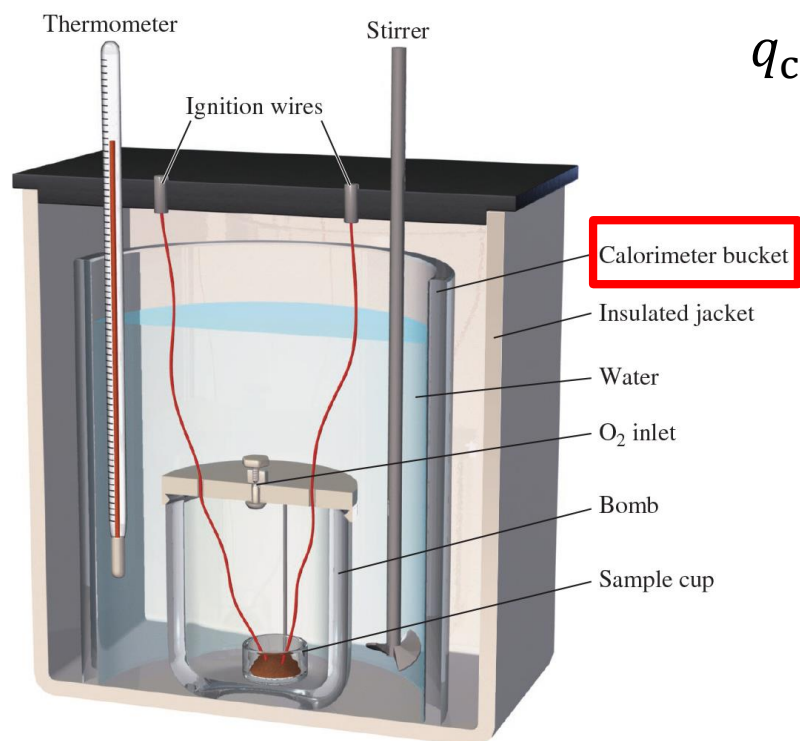
A 466-g sample of water is heated from 8.50°C to 74.60°C. Calculate the amount of heat absorbed (in kilojoules) by the water.

Solution Using Equation (6.12), we write

$$\begin{aligned} q &= ms\Delta t \\ &= (466 \text{ g}) (4.184 \text{ J/g} \cdot ^\circ\text{C}) (74.60^\circ\text{C} - 8.50^\circ\text{C}) \\ &= 1.29 \times 10^5 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} \\ &= 129 \text{ kJ} \end{aligned}$$

Constant-Volume Calorimetry

A constant-volume bomb calorimeter. The calorimeter is filled with oxygen gas at about 30 atm before it is placed in the bucket. The sample is ignited electrically, and the heat produced by the reaction can be accurately determined by measuring the temperature increase in the known amount of surrounding water.



$$q_{\text{sys}} = q_{\text{cal}} + q_{\text{rxn}}$$

$$= q_{\text{water}} + q_{\text{bomb}} + q_{\text{rxn}}$$

q_{cal} $\equiv$ the heat change for the calorimeter

q_{rxn} $\equiv$ the heat changes for the reaction

$$q_{\text{sys}} = 0 \quad (\because \text{No heat enters or leaves})$$

$$q_{\text{rxn}} = -q_{\text{cal}} = -(q_{\text{water}} + q_{\text{bomb}})$$

$$q_{\text{water}} = ms\Delta t$$

$$q_{\text{cal}} = C_{\text{cal}}\Delta t$$

$$q_{\text{bomb}} = C_{\text{bomb}}\Delta t$$

Reaction at Constant V

$$\Delta H \neq q_{\text{rxn}}$$

$$\Delta H \sim q_{\text{rxn}}$$

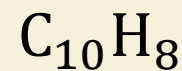
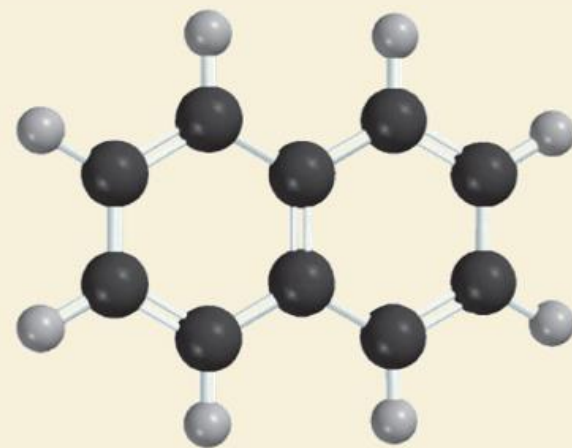
$$(\because q_V = \Delta U; q_p = \Delta H)$$

Constant pressure & ideal gas behavior:

$$\Delta U = \Delta H - RT\Delta n$$

Example 6.6

A quantity of 1.435 g of naphthalene (C_{10}H_8), a pungent-smelling substance used in moth repellents, was burned in a constant-volume bomb calorimeter. Consequently, the temperature of the water rose from 20.28°C to 25.95°C . If the heat capacity of the bomb plus water was $10.17\text{ kJ}/^\circ\text{C}$, calculate the heat of combustion of naphthalene on a molar basis; that is, find the molar heat of combustion.



Example 6.6 (1)

Solution

The heat absorbed by the bomb and water is equal to the product of the heat capacity and the temperature change.

Assuming no heat is lost to the surroundings,

$$q_{\text{cal}} = C_{\text{cal}}\Delta t = (10.17 \text{ kJ}/^{\circ}\text{C})(25.95^{\circ}\text{C} - 20.28^{\circ}\text{C}) = 57.66 \text{ kJ}$$

$$\therefore q_{\text{sys}} = q_{\text{cal}} + q_{\text{rxn}} = 0, q_{\text{rxn}} = -q_{\text{cal}} = -57.66 \text{ kJ}.$$

This is the heat released by the combustion of 1.435 g of C_{10}H_8 ; therefore, we can write the conversion factor as

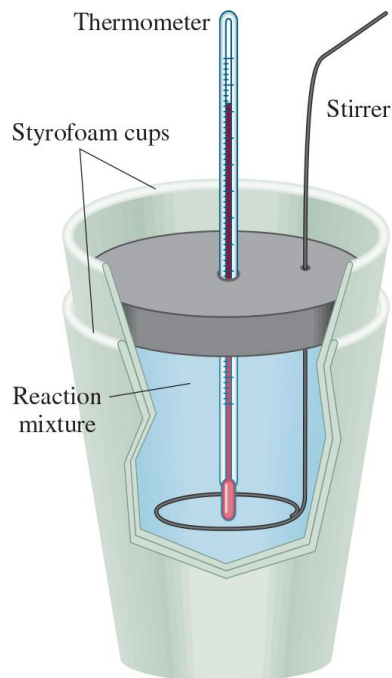
$$\frac{-57.66 \text{ kJ}}{1.435 \text{ g C}_{10}\text{H}_8}$$

The molar mass of naphthalene is 128.2 g, so the heat of combustion of 1 mole of naphthalene is

$$\begin{aligned} \text{molar heat of combustion} &= \frac{-57.66 \text{ kJ}}{1.435 \text{ g C}_{10}\text{H}_8} \times \frac{128.2 \text{ g C}_{10}\text{H}_8}{1 \text{ mol C}_{10}\text{H}_8} \\ &= -5.151 \times 10^3 \text{ kJ/mol} \end{aligned}$$

Constant-Pressure Calorimetry

A constant-pressure calorimeter made of two **Styrofoam coffee cups**. The outer cup helps to insulate the reacting mixture from the surroundings. Two solutions of known volume containing the reactants at the same temperature are carefully mixed in the calorimeter. The heat produced or absorbed by the reaction can be determined by measuring the temperature change.



Neglect the small heat capacity of the coffee cups.

This device measures the **heat of acid-base neutralization**, **heat of solution** and **heat of dilution**.

$$q_{\text{sys}} = q_{\text{water}} + q_{\text{cal}} + q_{\text{rxn}}$$

$$q_{\text{sys}} = 0 \quad (\because \text{No heat enters or leaves})$$

$$q_{\text{rxn}} = -(q_{\text{water}} + q_{\text{cal}})$$

$$q_{\text{water}} = ms\Delta t$$

$$q_{\text{cal}} = C_{\text{cal}}\Delta t$$

Reaction at Constant P

$$\Delta H = q_{\text{rxn}}$$

$$(\because q_V = \Delta U; q_P = \Delta H)$$

Table 6.3 Heats of Some Typical Reactions Measured at Constant Pressure

Type of Reaction	Example	ΔH (kJ/mol)
Heat of neutralization	$\text{HCl}(aq) + \text{NaOH}(aq) \longrightarrow \text{NaCl}(aq) + \text{H}_2\text{O}(l)$	-56.2
Heat of ionization	$\text{H}_2\text{O}(l) \longrightarrow \text{H}^+(aq) + \text{OH}^-(aq)$	56.2
Heat of fusion	$\text{H}_2\text{O}(s) \longrightarrow \text{H}_2\text{O}(l)$	6.01
Heat of vaporization	$\text{H}_2\text{O}(l) \longrightarrow \text{H}_2\text{O}(g)$	44.0*
Heat of reaction	$\text{MgCl}_2(s) + 2\text{Na}(l) \longrightarrow 2\text{NaCl}(s) + \text{Mg}(s)$	-180.2

Example 6.7

A lead (Pb) pellet having a mass of 26.47 g at 89.98° C was placed in a constant-pressure calorimeter of negligible heat capacity containing 100.0 mL of water. The water temperature rose from 22.50° C to 23.17° C. What is the specific heat of the lead pellet?

Solution Treating the calorimeter as an isolated system (no heat lost to the surroundings),

$$q_{\text{Pb}} + q_{\text{H}_2\text{O}} = 0 \Rightarrow q_{\text{Pb}} = -q_{\text{H}_2\text{O}}$$

The heat gained by the water: $q_{\text{H}_2\text{O}} = ms\Delta t$

where m and s are the mass and specific heat
and $\Delta t = t_{\text{final}} - t_{\text{initial}}$

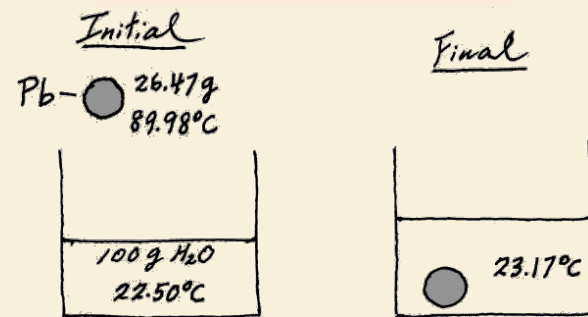
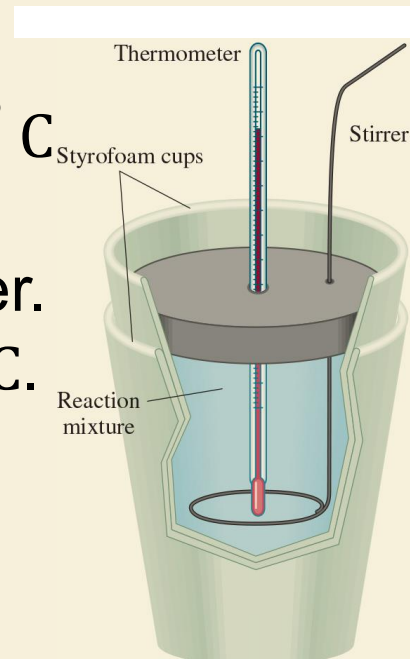
$$q_{\text{H}_2\text{O}} = (100.0 \text{ g})(4.184 \text{ J/g} \cdot ^\circ \text{C})(23.17^\circ \text{C} - 22.50^\circ \text{C}) = 280.3 \text{ J}$$

The heat lost by the lead pellet = the heat gained by the water,

$q_{\text{Pb}} = -280.3 \text{ J}$. Solving for the specific heat of Pb, we write

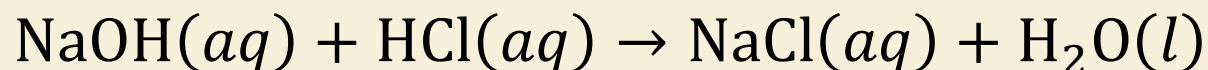
$$q_{\text{Pb}} = ms\Delta t \Rightarrow -280.3 \text{ J} = (26.47\text{g})(s)(23.17^\circ\text{C} - 89.98^\circ\text{C})$$

$$s = 0.158 \text{ J/g} \cdot ^\circ \text{C}$$



Example 6.8

A quantity of $1.00 \times 10^2 \text{ mL}$ of 0.500 M HCl was mixed with $1.00 \times 10^2 \text{ mL}$ of 0.500 M NaOH in a constant-pressure calorimeter of negligible heat capacity. The initial temperature of the HCl and NaOH solutions was the same, 22.50° C , and the final temperature of the mixed solution was 25.86° C . Calculate the heat change for the neutralization reaction on a molar basis:



Assume that the densities and specific heats of the solutions are the same as for water (1.00 g/mL and $4.184 \text{ J/g} \cdot ^\circ\text{C}$, respectively)

Example 6.8 (1)

Solution Assuming no heat is lost to the surroundings, $q_{\text{sys}} = q_{\text{soln}} + q_{\text{rxn}} = 0$, so $q_{\text{rxn}} = -q_{\text{soln}}$, where q_{soln} is the heat absorbed by the combined solution. Because the density of the solution is 1.00 g/mL, the mass of a 100-mL solution is 100 g.

$$\begin{aligned} q_{\text{soln}} &= ms\Delta t \\ &= (1.00 \times 10^2 \text{ g} + 1.00 \times 10^2 \text{ g})(4.184 \text{ J/g} \cdot ^\circ\text{C})(25.86^\circ\text{C} - 22.50^\circ\text{C}) \\ &= 2.81 \times 10^3 \text{ J} = 2.81 \text{ kJ} \end{aligned}$$

Because $q_{\text{rxn}} = -q_{\text{soln}}$, $q_{\text{rxn}} = -2.81 \text{ kJ}$.

From the molarities given, the number of moles of both HCl and NaOH in $1.00 \times 10^2 \text{ mL}$ solution is

$$\frac{0.500 \text{ mol}}{1\text{L}} \times 0.100 \text{ L} = 0.0500 \text{ mol}$$

Therefore, the heat of neutralization when 1.00 mole of HCl reacts with 1.00 mole of NaOH is

$$\text{heat of neutralization} = \frac{-2.81 \text{ kJ}}{0.0500 \text{ mol}} = -56.2 \text{ kJ/mol}$$

Enthalpy of Formation

Because there is no way to measure the **absolute value** of the enthalpy of a substance, must I measure the enthalpy change for every reaction of interest?

Establish an arbitrary scale with the **standard enthalpy of formation** (ΔH_f^0) as a reference point for all enthalpy expressions.

The standard enthalpy of formation of any **element** in its most stable form is **zero**.

1 atm and 25 °C:

$$\Delta H_f^0(\text{O}_2) = 0$$

$$\Delta H_f^0(\text{O}_3) = 142.2 \text{ kJ/mol}$$

$$\Delta H_f^0(\text{C, graphite}) = 0$$

$$\Delta H_f^0(\text{C, diamond}) = 1.90 \text{ kJ/mol}$$

Standard enthalpy of formation (ΔH_f^0) is the heat change that results when **1 mole** of a compound is formed from its **elements** at a pressure of **1 atm**.

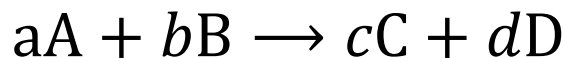
Some Standard Enthalpies of Formation

Table 6.4 Standard Enthalpies of Formation of Some Inorganic Substances at 25°C

Substance	$\Delta H_f^\circ(\text{kJ/mol})$	Substance	$\Delta H_f^\circ(\text{kJ/mol})$
Ag(<i>s</i>)	0	H ₂ O ₂ (<i>l</i>)	-187.6
AgCl(<i>s</i>)	-127.0	Hg(<i>l</i>)	0
Al(<i>s</i>)	0	I ₂ (<i>s</i>)	0
Al ₂ O ₃ (<i>s</i>)	-1669.8	HI(<i>g</i>)	25.9
Br ₂ (<i>l</i>)	0	Mg(<i>s</i>)	0
HBr(<i>g</i>)	-36.2	MgO(<i>s</i>)	-601.8
C(graphite)	0	MgCO ₃ (<i>s</i>)	-1112.9
C(diamond)	1.90	N ₂ (<i>g</i>)	0
CO(<i>g</i>)	-110.5	NH ₃ (<i>g</i>)	-46.3
CO ₂ (<i>g</i>)	-393.5	NO(<i>g</i>)	90.4
Ca(<i>s</i>)	0	NO ₂ (<i>g</i>)	33.85
CaO(<i>s</i>)	-635.6	N ₂ O(<i>g</i>)	81.56
CaCO ₃ (<i>s</i>)	-1206.9	N ₂ O ₄ (<i>g</i>)	9.66
Cl ₂ (<i>g</i>)	0	O(<i>g</i>)	249.4
HCl(<i>g</i>)	-92.3	O ₂ (<i>g</i>)	0
Cu(<i>s</i>)	0	O ₃ (<i>g</i>)	142.2
CuO(<i>s</i>)	-155.2	S(rhombic)	0
F ₂ (<i>g</i>)	0	S(monoclinic)	0.30
HF(<i>g</i>)	-271.6	SO ₂ (<i>g</i>)	-296.1
H(<i>g</i>)	218.2	SO ₃ (<i>g</i>)	-395.2
H ₂ (<i>g</i>)	0	H ₂ S(<i>g</i>)	-20.15
H ₂ O(<i>g</i>)	-241.8	Zn(<i>s</i>)	0
H ₂ O(<i>l</i>)	-285.8	ZnO(<i>s</i>)	-348.0

Enthalpy of Reaction

The **standard enthalpy of reaction** (ΔH_{rxn}^0) is the enthalpy of a reaction carried out at 1 atm.

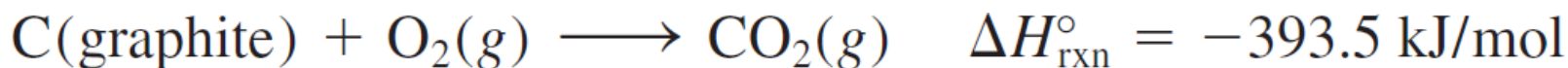


$$\Delta H_{\text{rxn}}^0 = [c\Delta H_f^0(C) + d\Delta H_f^0(D)] - [a\Delta H_f^0(A) + d\Delta H_f^0(B)]$$

$$\Delta H_{\text{rxn}}^0 = \sum n\Delta H_f^0(\text{products}) - \sum m\Delta H_f^0(\text{reactants})$$

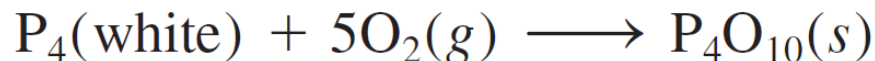
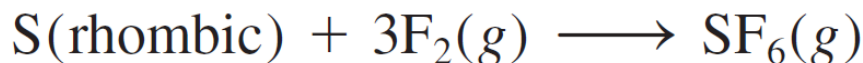
The Direct Method

This method of measuring ΔH_f^0 works for compounds that can be **readily synthesized** from their elements.



$$\begin{aligned}\Delta H_{\text{rxn}}^0 &= \Delta H_f^0(\text{CO}_2, g) - [\Delta H_f^0(\text{C, graphite}) + \Delta H_f^0(\text{O}_2, g)] \\ &= -393.5 \text{ kJ/mol}\end{aligned}$$

$$\Delta H_{\text{rxn}}^0 = \Delta H_f^0(\text{CO}_2, g) = -393.5 \text{ kJ/mol}$$



Enthalpy of Reaction (2)

The Indirect Method

Many compounds **cannot be directly synthesized** from their elements. In some cases, the reaction proceeds **too slowly**, or **side reactions** produce substances other than the desired compound.

$\Rightarrow \Delta H_f^0$ can be determined by an indirect approach, which is based on Hess's law of heat summation (Hess's law).

Hess's Law: When reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps.

$$\Delta H_{\text{rxn}}^0 = \sum n \Delta H_f^0 (\text{products}) - \sum m \Delta H_f^0 (\text{reactants})$$

Hess's law $\Rightarrow$ **H is a state function**

$\Rightarrow \Delta H$ depends only on the initial and final state (only on the nature of reactants and products).

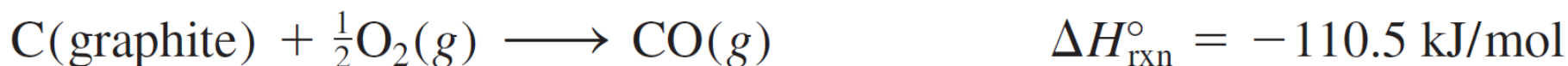
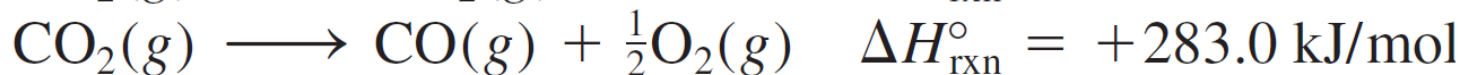
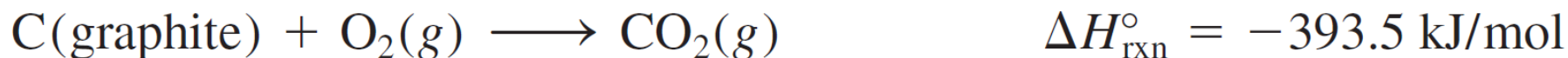
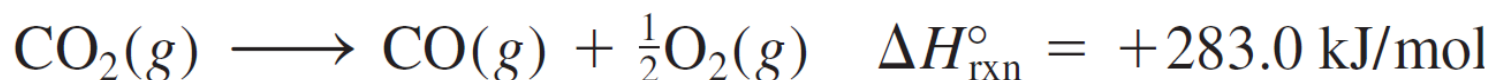
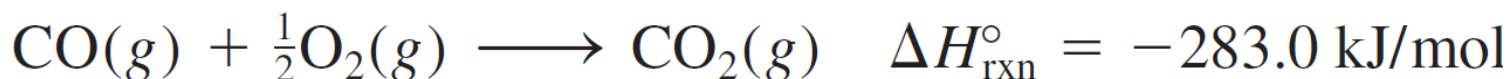
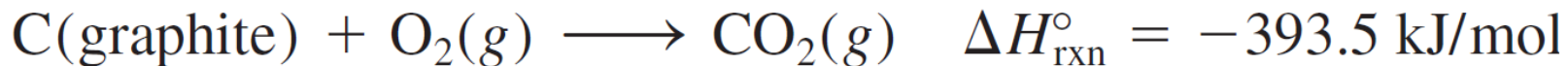
$\Rightarrow$ The enthalpy change would be the same whether the overall reaction takes place in one step or many steps.

Enthalpy of Reaction (3)

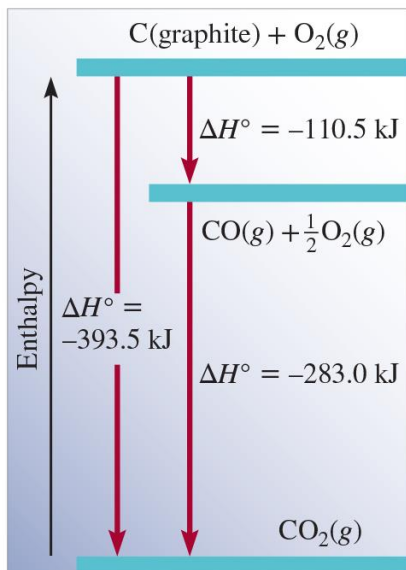
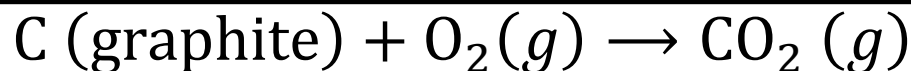
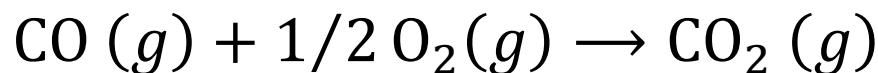
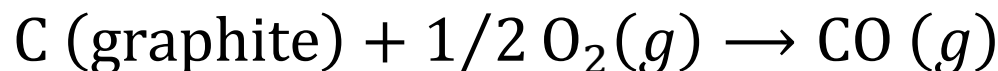
The standard enthalpy of formation of carbon monoxide (CO):



Table 6.4



$$\Rightarrow \Delta H_f^\circ(\text{CO}) = -110.5 \text{ kJ/mol}$$



The enthalpy change for the formation of 1 mole of CO_2 from graphite and O_2 can be broken down into two steps according to Hess's law.

Chemistry in Action:

How a Bombardier Beetle Defends Itself

(Survival techniques of insects and small animals)

Chameleons have developed the ability to change color to match their surroundings.

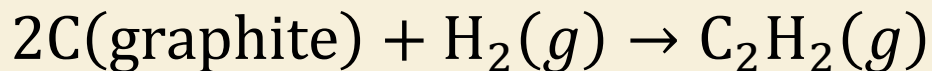
The butterfly *Limenitis* has evolved into a form that mimics the poisonous and unpleasant-tasting monarch butterfly (*Danaus*).

A less passive defense mechanism is employed by bombardier beetles (*Brachinus*), which repel predators with a “chemical spray.”

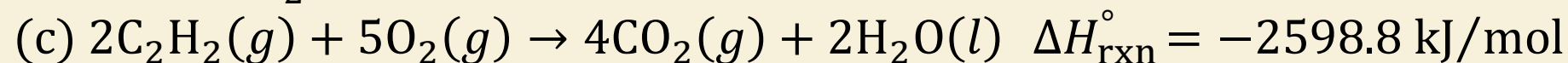
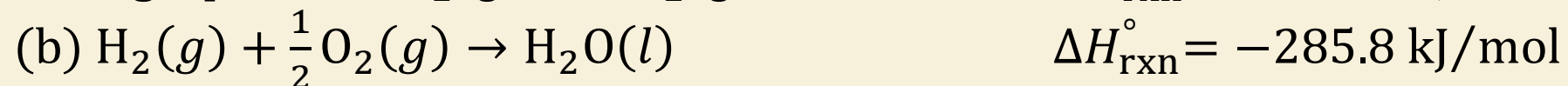


Example 6.9

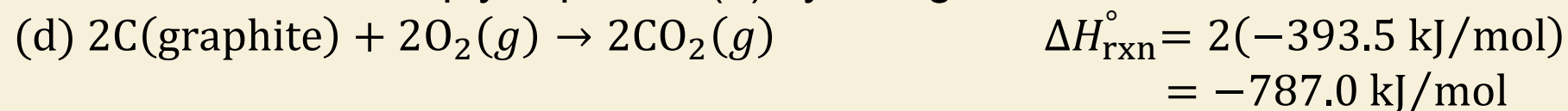
Calculate the standard enthalpy of formation of acetylene (C_2H_2) from its elements:



The equations for each step and the corresponding enthalpy changes are



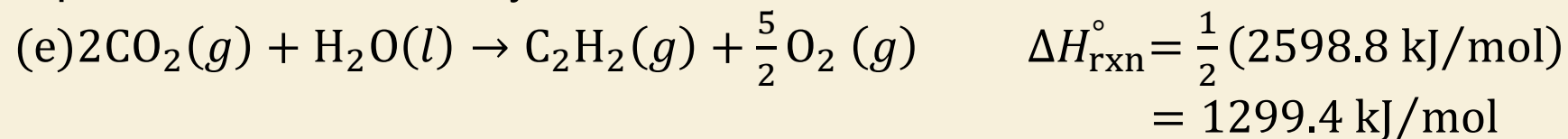
Solution Looking at the synthesis of C_2H_2 , we need 2 moles of graphite as reactant. So we multiply Equation (a) by 2 to get



Next, we need 1 mole of H_2 as a reactant and this is provided by Equation (b).

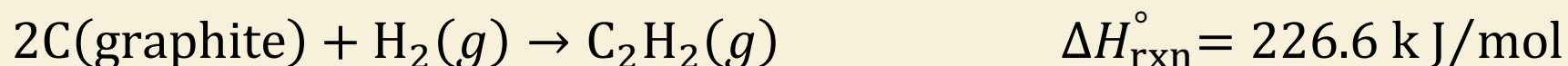
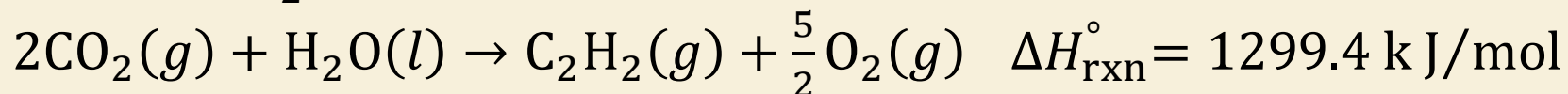
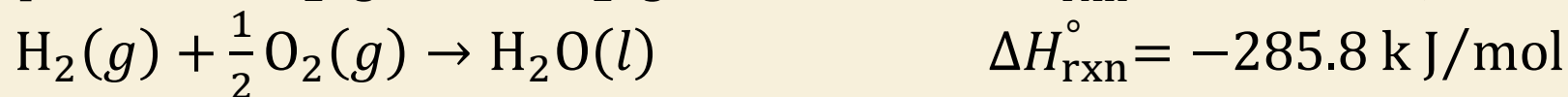
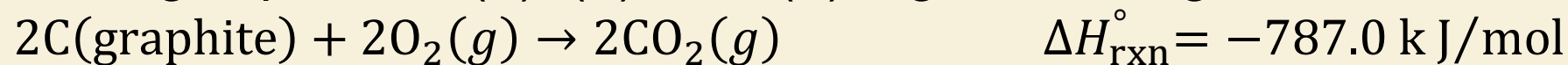
Last, we need 1 mole of C_2H_2 as a product.

Equation (c) has 2 moles of C_2H_2 as a reactant so we need to reverse the equation and divide it by 2:



Example 6.9 (1)

Adding Equations (d), (b), and (e) together, we get



Therefore,

$$\Delta H_{\text{f}}^{\circ} = 226.6 \text{ kJ/mol}$$

This value means that when 1mole of C_2H_2 is synthesized from 2 moles of $\text{C}(\text{graphite})$ and 1mole of H_2 , 226.6 kJ of heat are absorbed by the reacting system from the surroundings. Thus, this is an endothermic process.

Example 6.10

The thermite(鋁熱劑) reaction involves aluminum and iron(III) oxide: $2\text{Al}(s) + \text{Fe}_2\text{O}_3(s) \rightarrow \text{Al}_2\text{O}_3(s) + 2\text{Fe}(l)$

This reaction is highly exothermic and the liquid iron formed is used to weld metals. Calculate the heat released in kilojoules per gram of Al reacted with Fe_2O_3 . The ΔH_f° for $\text{Fe}(l)$ is 12.40 kJ/mol.



The molten iron formed in a thermite reaction is run down into a mold between the ends of two railroad rails. On cooling, the rails are welded together.

Example 6.10 (1)

Solution

Using the given ΔH_f° value for $\text{Fe}(l)$ and other ΔH_f° values in Appendix 3 and Equation (6.18), we write

$$\begin{aligned}\Delta H_{\text{rxn}}^\circ &= [\Delta H_f^\circ(\text{Al}_2\text{O}_3) + 2\Delta H_f^\circ(\text{Fe})] - [2\Delta H_f^\circ(\text{Al}) + \Delta H_f^\circ(\text{Fe}_2\text{O}_3)] \\ &= [(-1669.8 \text{ kJ/mol}) + 2(12.40 \text{ kJ/mol})] - [2(0) + (-822.2 \text{ kJ/mol})] \\ &= -822.8 \text{ kJ/mol}\end{aligned}$$

This is the amount of heat released for two moles of Al reacted. We use the following ratio

$$\frac{-822.8 \text{ kJ}}{2 \text{ mol Al}}$$

to convert to kJ/g Al.

The molar mass of Al is 26.98 g, so

$$\begin{aligned}\text{heat released per gram of Al} &= \frac{-822.8 \text{ kJ}}{2 \text{ mol Al}} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \\ &= 15.25 \text{ kJ/g}\end{aligned}$$

Heat of Solution and Dilution

The **heat of solution**, or **enthalpy of solution** (ΔH_{soln}) is the heat generated or absorbed when a certain amount of solute dissolves in a certain amount of solvent.

$$q_V = \Delta U; q_P = \Delta H \quad \Delta H_{\text{soln}} = H_{\text{soln}} - H_{\text{components (solute \& solvent)}}$$

Neither H_{soln} nor $H_{\text{components}}$ can be measured, but their difference, ΔH_{soln} , can be readily determined in a constant-pressure calorimeter.

$\Delta H_{\text{soln}} > 0$ for endothermic (heat-absorbing) processes

$\Delta H_{\text{soln}} < 0$ for exothermic (heat-generating) processes.

the heat of solution of a process in which an ionic compound is the solute and water is the solvent.

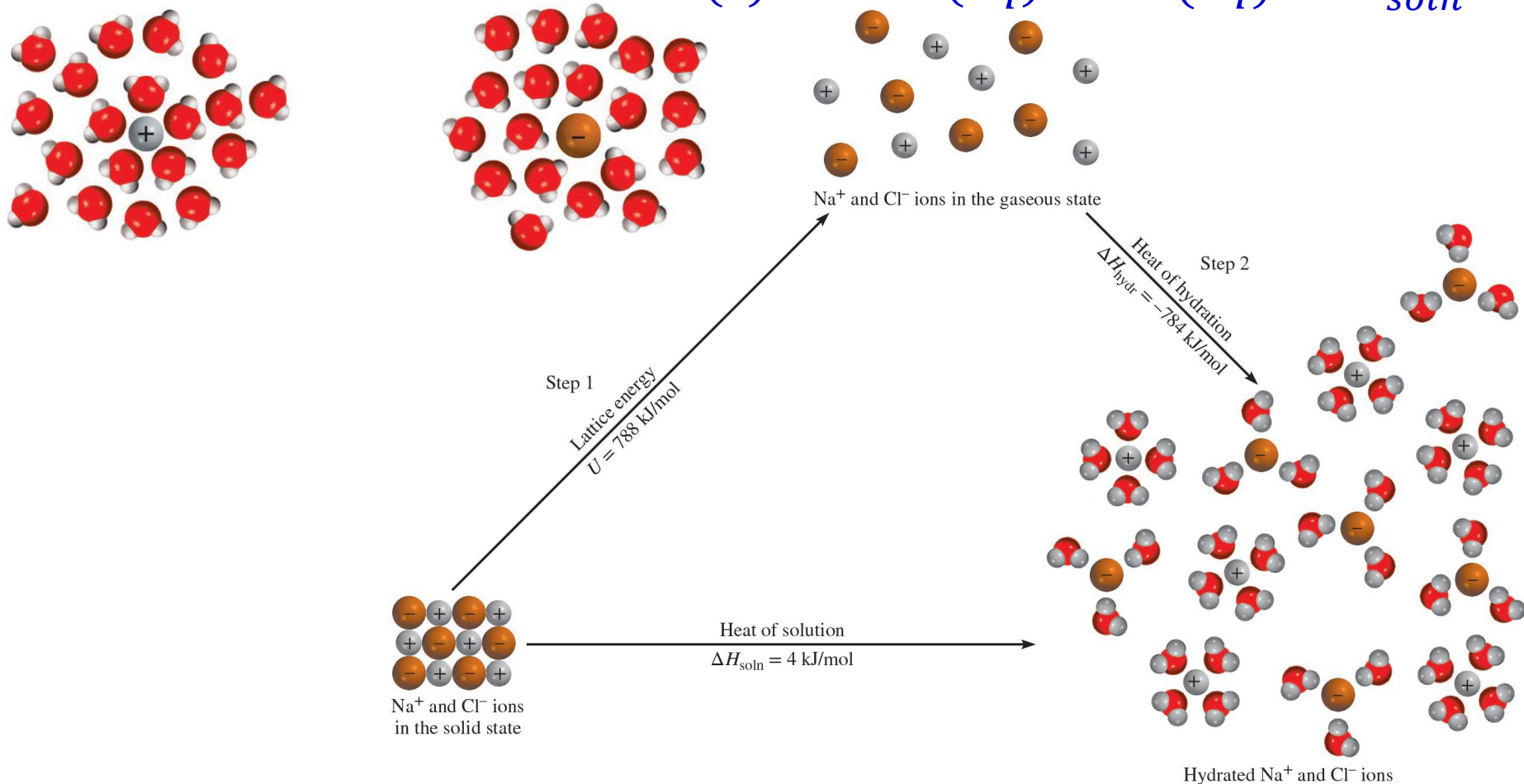
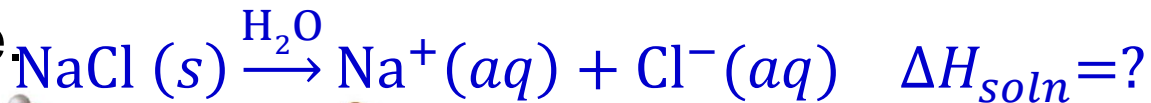
Table 6.5
Heats of Solution of
Some Ionic Compounds

Compound	ΔH_{soln} (kJ/mol)
LiCl	-37.1
CaCl ₂	-82.8
NaCl	4.0
KCl	17.2
NH ₄ Cl	15.2
NH ₄ NO ₃	26.6

Heat of Solution and Dilution (1)

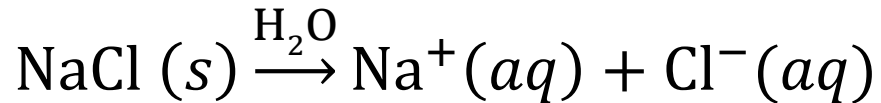
Consider the heat of solution of a process in which an ionic compound (NaCl) is the solute and water is the solvent.

Water molecules shield the ions (Na^+ and Cl^-) from each other and effectively reduce the electrostatic attraction that held them together in the solid state.

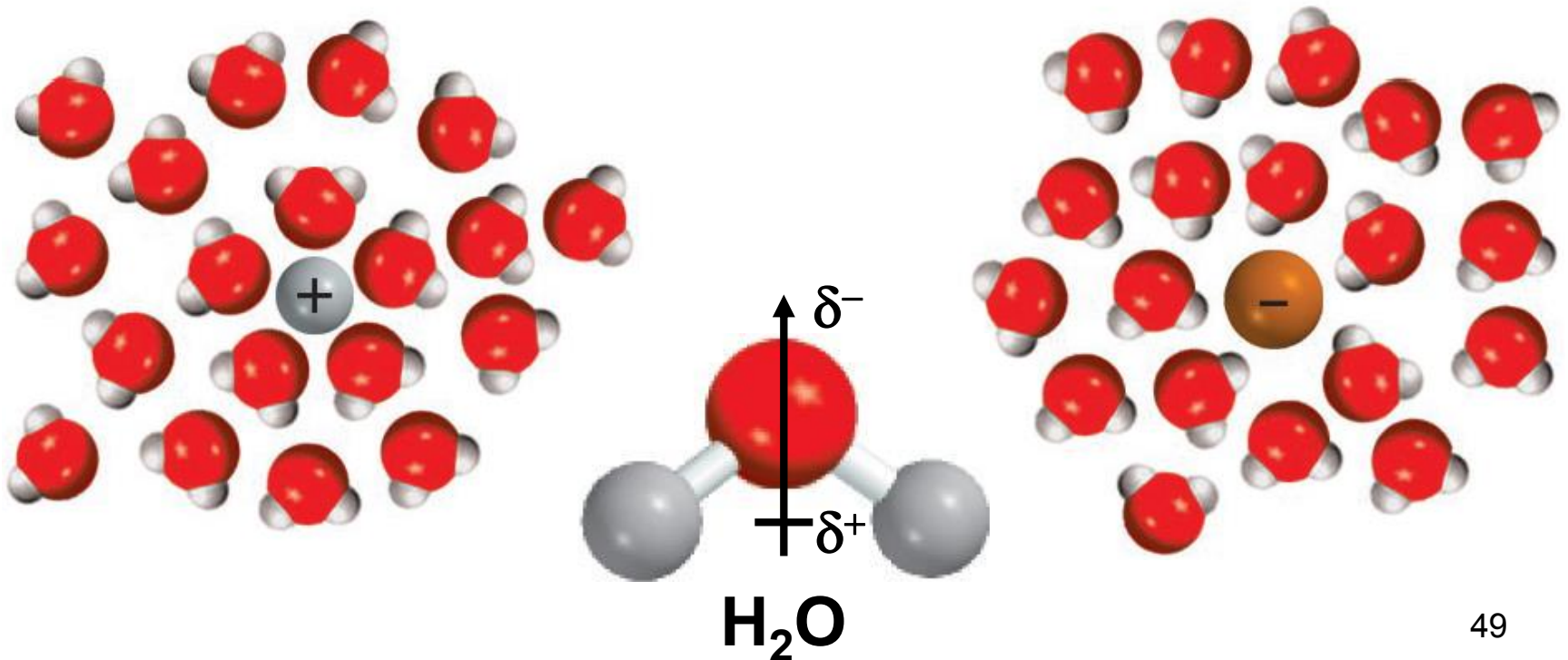


Hydration

Hydration is the process in which an ion is surrounded by water molecules arranged in a specific manner.

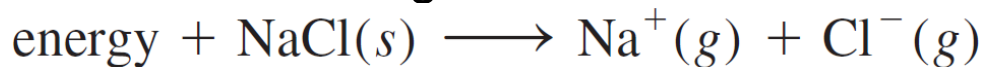


Each **Na⁺** ion is surrounded by a number of H₂O molecules orienting their **negative** poles toward the **cation**. Each **Cl⁻** ion is surrounded by H₂O molecules with their **positive** poles oriented toward the **anion**.



The Solution Process for NaCl

Step 1: Na^+ and Cl^- ions in the solid crystal are separated from each other and converted to the gaseous state:



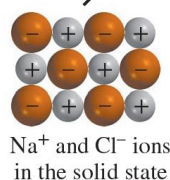
The energy required to completely separate **one mole** of a **solid ionic compound** into **gaseous ions** is called **lattice energy (U)**.

$$U_{\text{NaCl}} = 788 \text{ kJ/mol.}$$

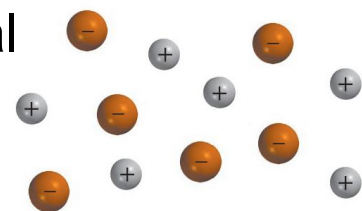
we would need to supply

788 kJ of energy to break 1 mole of solid NaCl into

1 mole of Na^+ ions and 1 mole of Cl^- ions.



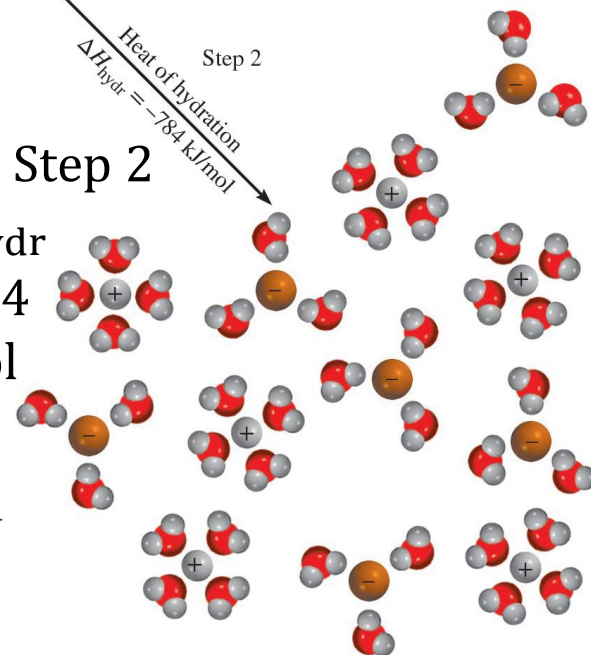
Step 1
Lattice energy
 $U = 788 \text{ kJ/mol}$



$$\begin{aligned} \Delta H_{\text{sol}} &= \text{Step 1} + \text{Step 2} \\ &= U + \Delta H_{\text{hydr}} \\ &= 788 - 784 \\ &= 4 \text{ kJ/mol} \end{aligned}$$

Heat of solution
 $\Delta H_{\text{soln}} = 4 \text{ kJ/mol}$

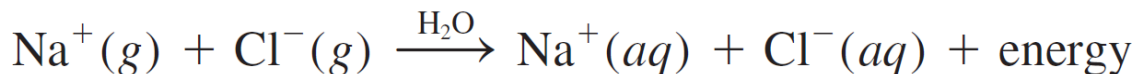
Step 2
Heat of hydration
 $\Delta H_{\text{hydr}} = -784 \text{ kJ/mol}$



Hydrated Na^+ and Cl^- ions

$$\Delta H_{\text{sol}} = U + \Delta H_{\text{hydr}}$$

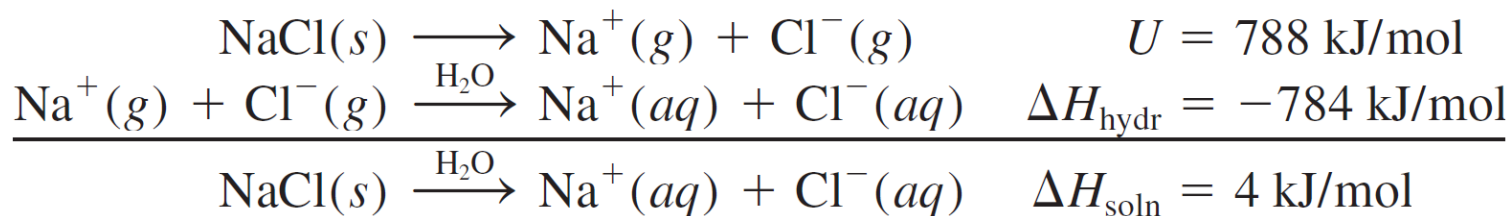
Step 2: the “gaseous” Na^+ and Cl^- ions enter the water and become hydrated



The enthalpy change associated with the hydration process is called the **heat of hydration**, ΔH_{hydr} (heat of hydration is a negative quantity for cations and anions). , $\Delta H_{\text{hydr}} \text{ of NaCl} = -784 \text{ kJ/mol.}$

The Solution Process for NaCl (1)

$$\Delta H_{sol} = U + \Delta H_{hydr}$$



The **heat of dilution** is the heat change associated with the dilution process.—If a certain solution process is **endothermic** and the solution is subsequently diluted, more heat will be absorbed by the same solution from the surroundings. The converse holds true for an **exothermic** solution process—more heat will be liberated if additional solvent is added to dilute the solution.

Because of its highly **exothermic heat of dilution**, concentrated sulfuric acid (H_2SO_4) poses a particularly hazardous problem if its concentration must be reduced by mixing it with additional water.

Concentrated H_2SO_4 is composed of 98 wt% acid and 2 wt% water. Never attempt to dilute the concentrated acid by adding water to it. The heat generated could cause the acid solution to boil and splatter. The recommended procedure is to add the concentrated acid slowly to the water (while constantly stirring).